

CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

This section identifies the functional dependencies of each entity in the **Max Online E-Commerce Database Management System** and analyzes the possible redundancy and data anomalies present in the database design. Functional dependencies describe the relationship between attributes, where one attribute uniquely determines another. Analyzing redundancy helps eliminate duplicate data, while identifying insertion, update, and deletion anomalies ensures data consistency and integrity. This analysis provides the foundation for database normalization and supports the design of an efficient, reliable, and well-structured relational database for the **Max Online E-Commerce Database Management System**.

4.2.1 CUSTOMER

Functional Dependency

- Customer_ID $\rightarrow$ First_Name, Last_Name, Email, Phone_Number, Date_of_Birth, Address

Redundancy Analysis

- Customer information is stored only once using **Customer_ID**, reducing duplicate customer records.

Data Anomalies

- **Insertion Anomaly:** Customer details cannot be stored without creating a customer record.

- **Update Anomaly:** Updating a customer's email, phone number, or address in multiple records may cause inconsistency

- **Deletion Anomaly:** Deleting a customer record removes all customer information and purchase history.

4.2.2 BRAND

Functional Dependency

- Brand_ID → Brand_Name, Country

Redundancy Analysis

- Brand details are stored only once and referenced by multiple products.

Data Anomalies

- **Insertion Anomaly:** Brand details cannot be stored without creating a brand record.

- **Update Anomaly:** Updating the brand name in multiple records may create inconsistency.

- **Deletion Anomaly:** Deleting a brand may affect all products associated with that brand.

4.2.3 CATEGORY

Functional Dependency

- Category_ID → Category_Name, Description

Redundancy Analysis

- Category information is stored only once and referenced by multiple products.

Data Anomalies

- **Insertion Anomaly:** A category cannot exist without a category record.
- **Update Anomaly:** Updating the category name in multiple records may create inconsistency.
- **Deletion Anomaly:** Deleting a category may affect all products belonging to that category.

4.2.4 PRODUCT

Functional Dependency

- Product_ID → Product_Name, Brand_ID, Category_ID, Price, Color, Strap_Material, Dial_Color, Water_Resistance, Stock

Redundancy Analysis

- Product information is stored only once and linked using **Product_ID**, preventing duplicate product records.

Data Anomalies

- **Insertion Anomaly:** Product details cannot be added without creating a product record.

- **Update Anomaly:** Updating product price, stock, or specifications in multiple records may cause inconsistencies.

- **Deletion Anomaly:** Deleting a product removes all information related to that product.

4.2.5 CART

Functional Dependency

- $\text{Cart_ID} \rightarrow \text{Customer_ID}, \text{Grand_Total}, \text{Total_Items}$

Redundancy Analysis

- Each customer has a separate shopping cart, reducing duplicate cart information.

Data Anomalies

- **Insertion Anomaly:** A cart cannot exist without a customer.
- **Update Anomaly:** Updating cart details in multiple places may cause inconsistency.
- **Deletion Anomaly:** Deleting a cart removes all items added by the customer.

4.2.6 ORDER

Functional Dependency

- $\text{Order_ID} \rightarrow \text{Customer_ID}, \text{Order_Date}, \text{Order_Status}, \text{Total_Amount}$

Redundancy Analysis

- Order information is stored once and linked to related order items and payments.

Data Anomalies

- **Insertion Anomaly:** An order cannot be created without customer information.

- **Update Anomaly:** Updating order status in multiple records may create inconsistency.
- **Deletion Anomaly:** Deleting an order removes the customer's purchase history.

4.2.7 ORDER_ITEM

Functional Dependency

- OrderItem_ID → Order_ID, Product_ID, Quantity, Unit_Price

Redundancy Analysis

- Each order item is uniquely identified, preventing duplicate product entries in an order.

Data Anomalies

- **Insertion Anomaly:** Order items cannot exist without an order.
- **Update Anomaly:** Updating quantity or unit price in multiple places may lead to inconsistency.
- **Deletion Anomaly:** Deleting an order item removes purchase details of that product.

4.2.8 PAYMENT

Functional Dependency

- Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

- Payment information is maintained separately, avoiding repeated payment details.

Data Anomalies

- **Insertion Anomaly:** Payment details cannot be stored without an order.

- **Update Anomaly:** Updating payment status in multiple records may create inconsistency.
- **Deletion Anomaly:** Deleting payment information removes transaction history.

4.2.9 REVIEW

Functional Dependency

- Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

- Customer reviews are stored separately to prevent duplicate feedback.

Data Anomalies

- **Insertion Anomaly:** Reviews cannot be added without a customer and product.
- **Update Anomaly:** Updating review information in multiple places may create inconsistency.
- **Deletion Anomaly:** Deleting a review removes valuable customer feedback.

4.2.10 SHIPMENT

Functional Dependency

- Shipment_ID → Order_ID, Delivery_Address, Shipment_Date, Delivery_Date, Shipment_Status

Redundancy Analysis

- Shipment details are stored separately, avoiding duplicate delivery information.

Data Anomalies

- **Insertion Anomaly:** Shipment information cannot exist without an order.
- **Update Anomaly:** Updating shipment status in multiple records may create inconsistency.
- **Deletion Anomaly:** Deleting shipment information removes delivery history.

4.2.11 WARRANTY

Functional Dependency

- Warranty_ID → Product_ID, Warranty_Period, Start_Date, End_Date

Redundancy Analysis

- Warranty information is maintained separately for each product, reducing duplicate warranty records.

Data Anomalies

- **Insertion Anomaly:** Warranty details cannot be stored without an associated product.
- **Update Anomaly:** Updating warranty information in multiple records may create inconsistency.
- **Deletion Anomaly:** Deleting warranty information removes product warranty history.

4.3 CONCLUSION

The functional dependency analysis of the **Max Online E-Commerce Database Management System** demonstrates that each entity has well-defined relationships between its attributes, ensuring that every non-key attribute is fully functionally dependent on its primary key. The analysis of redundancy and data anomalies highlights the importance of eliminating duplicate data and preventing insertion, update, and deletion anomalies through proper database design. By identifying these functional

dependencies and applying normalization techniques, the database achieves improved data integrity, consistency, and efficiency. This analysis provides a strong foundation for developing a reliable, scalable, and well-structured relational database that supports the effective management of **customers, brands, categories, products, shopping carts, orders, order items, payments, reviews, shipments, and warranties** in the **Max Online E-Commerce Database Management System**.